



Human-AI Collaboration in a Student Discussion Forum

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ABSTRACT

The recent public releases of AI tools such as ChatGPT have forced computer science educators to reconsider how they teach. These tools have demonstrated considerable ability to generate code and answer conceptual questions, rendering them incredibly useful for completing CS coursework. While overreliance on AI tools could hinder students' learning, we believe they have the potential to be a helpful resource for both students and instructors alike. We propose a novel system for instructor-mediated GPT interaction in a class discussion board. By automatically generating draft responses to student forum posts, GPT can help Teaching Assistants (TAs) respond to student questions in a more timely manner, giving students an avenue to receive fast, quality feedback on their solutions without turning to ChatGPT directly. Additionally, since they are involved in the process, instructors can ensure that the information students receive is accurate, and can provide students with incremental hints that encourage them to engage critically with the material, rather than just copying an AI-generated snippet of code. We utilize Piazza—a popular educational forum where TAs help students via text exchanges—as a venue for GPT-assisted TA responses to student questions. These student questions are sent to GPT-4 alongside assignment instructions and a customizable prompt, both of which are stored in editable instructor-only Piazza posts. We demonstrate an initial implementation of this system, and provide examples of student questions that highlight its benefits.

CCS CONCEPTS

• Applied computing → Computer-assisted instruction; • Social and professional topics → Computing education; • Computing methodologies → Artificial intelligence; • Human-centered computing → Interactive systems and tools.

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1 INTRODUCTION & BACKGROUND

1.1 AI in the Classroom

In the past three years, large language model-based tools such as ChatGPT and GitHub Copilot have upended computer science education, as instructors scramble to adapt their courses to a world in which assignments can be automatically completed with a click of a button [5–7]. Preliminary surveys have indicated that educators are divided in their beliefs about the best path forward, with some instructors wanting to incorporate AI tools into their courses and others preferring to ban and avoid them [6].

Instructors who wish to adopt models like ChatGPT cite a number of potential benefits, including faster and more personalized support for students and assistance with time-consuming tasks for instructors. On the other hand, opponents of this idea worry that it could remove opportunities for students to explore and discover, believing that an overreliance on AI tools could hinder learning and allow students to complete assignments without absorbing the course material [2, 6, 9, 10]. Other concerns shared by instructors and students alike include the possibility of receiving incorrect or outdated information and the inability to ask nuanced, course-specific questions [9, 10].

1.2 Automated Teaching Assistants

A clear application of AI in CS courses is to automate the duties of Teaching Assistants (TAs), especially with regards to answering students' questions in online class discussion boards. One early work in this direction is Jill Watson, a virtual teaching assistant that uses IBM Watson to automatically answer forum questions for a knowledge-based artificial intelligence course [4]. Jill has been able to achieve over ninety percent accuracy on the questions she answers, but is mostly limited to routine, administrative topics.

Other works have investigated the capacity of large language models to identify and explain errors in student code. Balse et al. finds that GPT-3.5 Turbo is capable of generating explanations that are comparable in quality to those written by human TAs for an introductory programming course, and the vast majority of such explanations correctly identified at least one logical error. However, half of the explanations contained at least one incorrect statement, and a third of them failed to mention at least one known error [1]. The authors suggest that while trained human TAs may be able to extract value from these generated explanations—by using them as a starting point and to identify where the issues lie in a student's code—they are not always sophisticated enough to be presented to students as-is [1].

2 MEDIATED GPT AGENT FOR PIAZZA

We seek to alleviate the issues outlined above by introducing a novel system for instructor-mediated GPT interaction in a class discussion board. Our tool is built upon Piazza, a popular course forum that is already being used to facilitate student-instructor interaction in thousands of universities [8]. It utilizes GPT-4 to automatically generate draft replies to student help requests under the TA's Piazza account. Thus, when the TA logs into Piazza to answer a student's question, they will already have a rough response drafted that they can further edit and check for correctness before posting to the student. By integrating our tool with Piazza, we incorporate it into the existing workflow of students and instructors, thereby encouraging its use among students and giving TAs the flexibility to answer questions quickly from any device.

The tool is implemented in Java and uses the latest version of GPT-4 Turbo to generate draft responses to student questions based on a custom prompt (Figure 1). Post tags are used to determine which assignment write-up is relevant to a student's question, and these assignment instructions are sent to the GPT model alongside the student's question and the prompt. All data used and/or generated by the tool is stored inside private, instructor-only Piazza posts, enabling on-the-fly editing and eliminating the need for locally-stored files.

We demonstrate the utility of our tool by applying it to a real exercise used in an advanced undergraduate Programming Language Concepts course. The exercise served as a free response exam question about concurrency in Java, specifically with regards to race conditions and the synchronized keyword. Figure 2 contains code for an `incrementCounter()` method that is being run by multiple threads, while Figure 3 illustrates a sample forum post in which a student describes their current thought process and asks for assistance in figuring out next steps. The student's question specifically references part (a) of the assignment, which reads: "Give the output of this program, in which none of the methods is synchronized. Be sure to explain your answer". Since the lack of synchronization means `incrementCounter()` is never locked, the output should be 1 due to interleaving threads (the counter starts at 0).

The draft explanation for this question generated by our tool can be found in Figure 4, and Figure 5 contains a revised version of this explanation after edits were made by a human TA. Although the instructor consolidated some points and introduced a few new ideas, the fully AI-generated explanation is quite thorough, and it succeeds in addressing the specific aspects of the assignment that the student had cited as confusing. Routinely generating explanations that are this in-depth from scratch would take a considerable amount of time and thought for a TA, whereas reviewing them and correcting them can be performed relatively quickly.

In addition to drafting a reply to the student's post, our tool creates a corresponding instructor-only thread which contains the original student question, the full response given by GPT, and a link to the student's post. This way, if the TA makes substantial edits to the drafted reply, they can go back and reference the original GPT response. Similarly, the drafted reply on the student's post contains a footer with a link to this instructor-only post (see Figure 5), allowing for easy navigation between the two threads.

When this same exercise was administered for a class of real students, around 30 percent of them failed to provide the correct program output for part (a). Of those who had incorrect answers, nearly all also failed to give a correctly reasoned explanation for their answer. This is a situation in which access to instructor-mediated GPT could have been extremely helpful: allowing students to promptly receive new explanations of key concepts they are struggling with may encourage them to actively think through the problem and eventually reach the correct answer on their own.

Furthermore, our tool provides several notable benefits when compared with other solutions. Existing automated TAs such as Jill Watson—or for that matter, base ChatGPT—provide students with the full answer right away, lessening the need for students to engage with the material or understand the intermediate steps involved in a problem. In contrast, our tool is instructed to use a step-by-step natural language approach in its explanations, breaking the problem down into a series of smaller problems that the student knows how to solve. Additionally, other tools are capable of giving information that is incorrect, outdated, out-of-scope, or otherwise confusing or misleading for students, an issue which TA supervision eliminates.

3 DISCUSSION & FUTURE WORKS

We introduce here a tool for facilitating TA-mediated interactions with GPT over Piazza. Our tool is designed to reduce the workload of TAs, allowing them to spend less time answering common, surface-level questions and more time engaging with students that need in-depth, personalized help. We hope that it will have the added benefit of reducing student ChatGPT use, since students may be less inclined to turn to AI tools on their own if they can receive quality feedback to forum posts quickly. Although it is nearly impossible to completely police students' ChatGPT use, we believe that our tool can provide the foundation for a healthy and thoughtful integration of AI into the classroom, even for courses in which direct student access to GPT is forbidden by the honor code.

A potential area of concern is that such a tool may be mostly useful for entry-level programming courses, where the issues encountered by students have simple solutions that are readily found online. However, the example we have provided is from an assignment on concurrency—an advanced topic from an advanced programming course—and in general, the tool has the potential to resolve the types of complex coding issues that can occur in advanced classes. In a similar vein, some student questions may require more course-specific contextual knowledge to solve than others. We have thus far found it suitable to simply include the entire assignment instructions in the GPT prompt, but we are interested in implementing mechanisms to retrieve other relevant classroom material to send to GPT.

In the future, we wish to investigate the characteristics of drafted explanations. We plan to conduct an experiment with human judges in order to determine the perceived quality of AI-generated responses relative to responses written by human TAs. We intend to measure the average number and degree of changes made to drafted responses by TAs before sending them out to students and whether students are generally satisfied with the feedback they receive. Furthermore, we hope to test the efficacy of our tool in comparison to existing approaches such as unmediated ChatGPT,

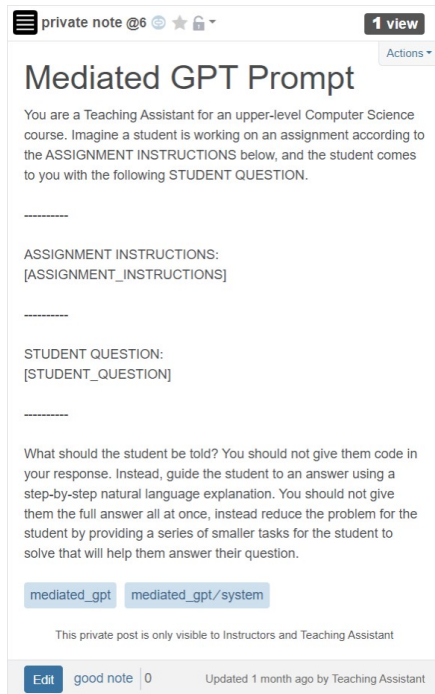


Figure 1: A screen capture of the Piazza post containing the GPT-4 prompt template.

```
//synchronized
static void incrementCounter() {
    int aRegister = CounterIncrementer.counter;
    System.out.println("Loaded:" + aRegister);
    aRegister++;
    ThreadSupport.sleep(10);
    // The above sleep call will block the current
    // thread for 10 ms and let some other thread
    // execute before the call unblocks
    CounterIncrementer.counter = aRegister;
    System.out.println("Saved:" + aRegister);
}
```

Figure 2: A snippet of Java code from a set of sample assignment instructions.

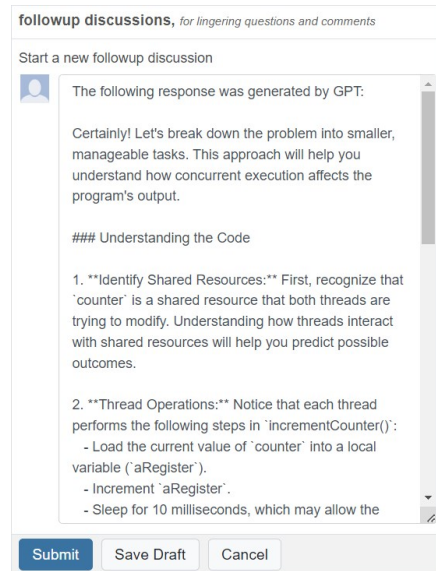


Figure 4: The reply automatically drafted by our tool in response to Figure 3's question.

Question about threads

I'm working on the threads assignment and I'm stuck on part (a). I think different threads will execute at the same time, but I can't figure out how to determine what the actual output will be. Any tips?

mediated_gpt mediated_gpt/help_needed
example_assignment example_assignment/q1

Figure 3: A sample student question regarding the exercise involving Figure 2's code.

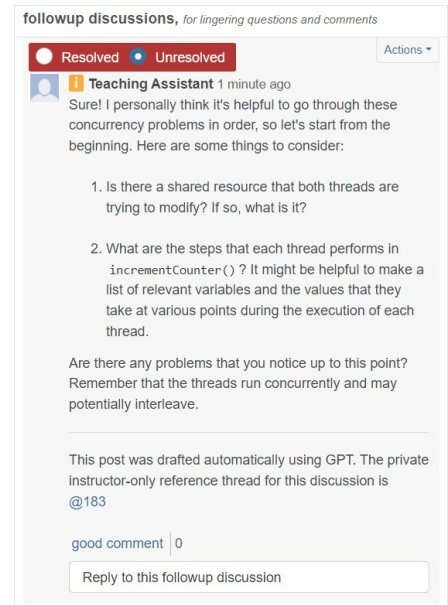


Figure 5: The reply actually sent out to the student after the draft provided in Figure 4 was edited by a TA.

unassisted human TAs, and other automated assistants. There are a number of metrics that should be considered in this evaluation, including the usability, effectiveness, and efficiency from the TA perspective, and the helpfulness, correctness, and timeliness from the student perspective.

While our tool facilitates human mediation of AI, it does little in the way of automated mediation. One work in this direction is George and Dewan [3], which introduces the NotebookGPT tool for programmatically mediating student-GPT interactions inside a Jupyter Notebook environment. Though it provides less control than our tool due to its automated nature, NotebookGPT allows students to receive real-time mediated feedback without having to make a forum post or wait for an instructor to respond. Taking inspiration from this work, we plan to integrate our tool with existing test suites to automatically generate private forum posts for students that include their code, failed test cases, and relevant assignment information. We believe that instructor-mediated AI has tremendous potential to enhance computer science education,

and we intend to continue improving our tool in order to increase TA efficiency, promote student learning, and gain insights about the capability of GPT models.

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